Module IV — Magnetic & dielectric properties

Derivations, step by step

Langevin theory of diamagnetism

8 marks

An electron (charge $-e$, mass m) goes round a nucleus $+Ze$ in a circle of radius ρ with angular frequency ω_0. A field B is applied at right angles to the orbit. N atoms per unit volume, Z electrons each.

BEFORE YOU START: WHAT THIS USES

$Ze,\ -e$	nuclear charge and electron charge; Z electrons per atom: where an atomic moment comes from
$m\omega_0^2\rho = \frac{Ze^2}{4\pi\varepsilon_0\rho^2}$	Newton's law with Coulomb attraction as the centripetal force: the first step
$F = qv \times B$	Lorentz force on a moving charge; for $v \perp B$ its size is qvB: used in the second phase
$\mu = IA$	magnetic moment of a current loop: where an atomic moment comes from
$\chi = \mu_0 M/B$	susceptibility and magnetisation $M = N\mu$: magnetic terminology

START

$$m\omega_0^2\rho = \frac{Ze^2}{4\pi\varepsilon_0\rho^2}$$

RESULT

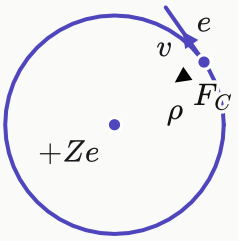
$$\chi_{\text{dia}} = -\frac{\mu_0 N Z e^2}{6m}\langle r^2\rangle$$

WITHOUT THE FIELD

$$m\omega_0^2\rho = \frac{Ze^2}{4\pi\varepsilon_0\rho^2}$$

The Coulomb attraction supplies the centripetal force.

Coulomb force $= m\omega_0^2\rho$



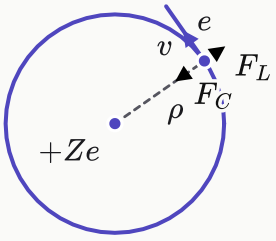
SWITCH ON B

$$m\omega^2\rho = \frac{Ze^2}{4\pi\varepsilon_0\rho^2} - e\rho\omega B$$

Add the Lorentz force $evB = e\rho\omega B$ (radially out for this sense of motion)

The Lorentz force on a charge moving with velocity $\mathbf{v}$ in a field $\mathbf{B}$ is $q\mathbf{v} \times \mathbf{B}$. Here $\mathbf{v} \perp \mathbf{B}$, so its size is evB with $v = \rho\omega$, and it is along the radius. For this direction of motion it points outwards and weakens the pull; for the opposite sense it adds to it.

The Lorentz force F_L acts along the radius



$$m\omega^2\rho = m\omega_0^2\rho - e\rho\omega B$$

The Coulomb term is $m\omega_0^2\rho$ (first line)

$$\omega^2 + \frac{eB}{m}\omega - \omega_0^2 = 0$$

Divide by $m\rho$ and collect

A quadratic in ω.

LARMOR FREQUENCY

$$\omega = -\frac{eB}{2m} \pm \sqrt{\left(\frac{eB}{2m}\right)^2 + \omega_0^2}$$

Quadratic formula

$$\omega \approx \omega_0 - \frac{eB}{2m}$$

Laboratory fields: $eB/2m \ll \omega_0$

The orbit frequency changes by $\omega_L = eB/2m$, the Larmor frequency (Larmor's theorem). Electrons with moments along B slow down; those against it speed up.

Check. Take $eB/2m \ll \omega_0$. What does the square root become?

- ✓ about ω_0, so $\omega \approx \omega_0 - eB/2m$
- about $eB/2m$
- zero
- ω_0^2

$\sqrt{(eB/2m)^2 + \omega_0^2} \approx \omega_0$ when ω_0 is much the larger.

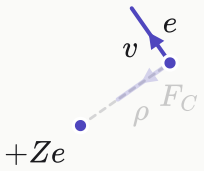
EXTRA CURRENT, EXTRA MOMENT

$$I = -e \cdot \frac{1}{2\pi} \cdot \frac{eB}{2m} = -\frac{e^2 B}{4\pi m}$$

Current = charge × change in revolutions per second

A charge going round f times per second carries a current $I = qf$. The change in frequency is $\Delta f = \Delta\omega/2\pi = \frac{1}{2\pi}\frac{eB}{2m}$, and it is an extra circulating current.

> The extra circulation is a current loop



$$\mu = I\pi\rho^2 = -\frac{e^2 B}{4m}\rho^2$$

multiply: $\pi\rho^2$

Moment of a current loop = current × area (atomic moments). It is negative: it opposes B (Lenz's law).

ALL THE ELECTRONS

$$\mu = -\frac{Ze^2 B}{4m}\langle\rho^2\rangle$$

Z electrons with randomly oriented orbits: use the mean $\langle\rho^2\rangle$

ρ is now the distance from the field axis: $\langle\rho^2\rangle = \langle x^2\rangle + \langle y^2\rangle$.

$$\langle\rho^2\rangle = \frac{2}{3}\langle r^2\rangle$$

Spherical atom: $\langle x^2\rangle = \langle y^2\rangle = \langle z^2\rangle = \frac{1}{3}\langle r^2\rangle$

Check. For a spherical charge cloud, what fraction of $\langle r^2\rangle$ is $\langle x^2\rangle + \langle y^2\rangle$?

- ✓ $2/3$
- $1/3$
- $1/2$
- 1

Each of the three directions carries a third of $\langle r^2\rangle$; two of them are perpendicular to B.

$$\mu = -\frac{Ze^2 B}{6m}\langle r^2\rangle$$

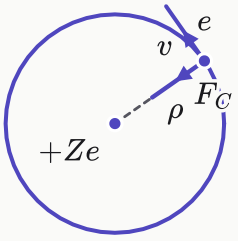
substitute: $\langle\rho^2\rangle = \frac{2}{3}\langle r^2\rangle$

SUSCEPTIBILITY

$$\chi_{\text{dia}} = \frac{\mu_0 N\mu}{B} = -\frac{\mu_0 NZe^2}{6m}\langle r^2\rangle$$

$$M = N\mu,\ \chi = M/H = \mu_0 M/B$$

Negative, small (He: -1.9×10^{-6}) and independent of temperature, as observed. Nothing here needs a permanent moment, so every material is diamagnetic underneath. The quantum theory (diamagnetism) gives exactly the same result.



Langevin theory of paramagnetism and the Curie law

8 marks

N atoms per unit volume, each with a permanent moment μ, free to turn, not interacting. θ: angle between a moment and $\mathbf{B}$. k: Boltzmann's constant.

BEFORE YOU START: WHAT THIS USES

μ	permanent magnetic moment of an atom: where an atomic moment comes from
$E = -\mu B\cos\theta$	energy of a moment in a field: derived in the first steps below
$e^{-E/kT}$	Boltzmann factor: the classical form of the Fermi function, Boltzmann tail
$2\pi\sin\theta\,d\theta$	solid angle of a ring: derived in the step with that label
$\chi = \mu_0 M/B$	susceptibility: magnetic terminology

START

$$\tau = \mu B\sin\theta$$

RESULT

$$\chi_{\text{para}} = \frac{\mu_0 N\mu^2}{3kT} = \frac{C}{T}$$

ENERGY OF A MOMENT

$$\tau = \mu B\sin\theta$$

Torque on the moment (this is what "moment" means: torque per unit field at $90°$)

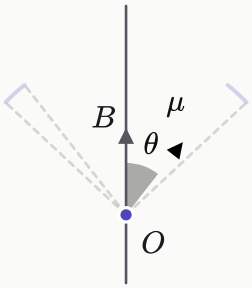
The torque turns the moment towards $\mathbf{B}$, reducing θ.

$$E = -\mu B \cos\theta$$

Energy = work done against the torque, measured from $\theta = 90°$

The field tries to line the moments up (lowest energy at $\theta = 0$); thermal agitation keeps them random.

Moment μ at angle θ to B



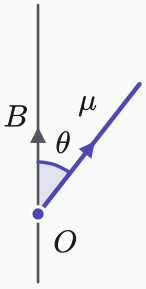
HOW MANY AT EACH ANGLE

$$dn = A\, e^{\mu B \cos\theta / kT}\, 2\pi \sin\theta\, d\theta$$

Boltzmann factor × solid angle of the ring between θ and $\theta + d\theta$

The fraction of moments with energy E is proportional to $e^{-E/kT}$ (Boltzmann; the same exponential as the tail of the Fermi function in the electron derivation). Moments pointing between θ and $\theta + d\theta$ fill a ring on the unit sphere: circumference $2\pi \sin\theta$ times width $d\theta$, so the solid angle is $2\pi \sin\theta\, d\theta$.

The ring between θ and $\theta + d\theta$



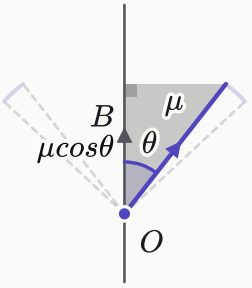
AVERAGE COMPONENT ALONG B

$$\mu \cos\theta$$

Each moment contributes its component along B

The components across the field cancel in pairs.

Component along $B = \mu \cos\theta$



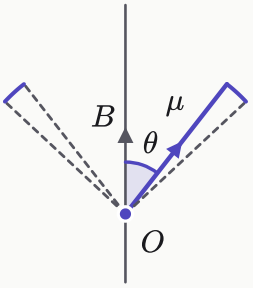
Check. In the right triangle formed by the moment and the field axis, what is the component of μ along B?

- ✓ $\mu \cos\theta$
- $\mu \sin\theta$
- $\mu \tan\theta$
- $\mu / \cos\theta$

The component is the side next to θ: hypotenuse μ times $\cos\theta$.

$$M = N\mu \frac{\int_0^\pi \cos\theta\, e^{x\cos\theta} \sin\theta\, d\theta}{\int_0^\pi e^{x\cos\theta} \sin\theta\, d\theta}$$

Average over the distribution, with $x = \mu B / kT$



DO THE INTEGRALS

$$M = N\mu \frac{\int_{-1}^{1} y\, e^{xy}\, dy}{\int_{-1}^{1} e^{xy}\, dy}$$

substitute: $y = \cos\theta,\ dy = -\sin\theta\, d\theta$

$$\int_{-1}^{1} e^{xy}\, dy = \frac{e^x - e^{-x}}{x}$$

The bottom integral

$$\int_{-1}^{1} y\, e^{xy}\, dy = \frac{e^x + e^{-x}}{x} - \frac{e^x - e^{-x}}{x^2}$$

The top integral, by parts: $u = y,\ dv = e^{xy} dy$

$\left[\frac{y\, e^{xy}}{x}\right]_{-1}^{1} - \frac{1}{x}\int_{-1}^{1} e^{xy} dy$.

$$M = N\mu\left[\frac{e^{x}+e^{-x}}{e^{x}-e^{-x}} - \frac{1}{x}\right]$$

Divide the top by the bottom

$$M = N\mu\left[\coth x - \frac{1}{x}\right] = N\mu\, L(x)$$

Langevin function $L(x) = \coth x - 1/x$

Large x (strong field or low T): $L \to 1$, $M \to N\mu$, saturation: every moment aligned.

ORDINARY FIELDS

$$\coth x = \frac{1 + \frac{x^2}{2} + \dots}{x\left(1 + \frac{x^2}{6} + \dots\right)} \approx \frac{1}{x} + \frac{x}{3}$$

Series for small x: $\cosh x = 1 + \frac{x^2}{2}$, $\sinh x = x + \frac{x^3}{6}$

$\frac{1}{x}\left(1+\frac{x^2}{2}\right)\left(1-\frac{x^2}{6}\right) \approx \frac{1}{x}\left(1+\frac{x^2}{3}\right)$.

$$L(x) \approx \frac{x}{3}$$

For $x \ll 1$: $\coth x \approx \frac{1}{x} + \frac{x}{3}$

At room temperature in a laboratory field $\mu B \ll kT$, $x \sim 10^{-3}$.

Check. For small x, $\coth x \approx \frac{1}{x} + \frac{x}{3}$. So $L(x) = \coth x - \frac{1}{x} \approx$?

- ✓ $x/3$
- $1/x$
- 1
- $3x$

The $1/x$ terms cancel and $x/3$ is left: the slope of L at the origin.

$$M = \frac{N\mu^2 B}{3kT}$$

substitute: $L = x/3$, $x = \mu B/kT$

$$\chi_{\text{para}} = \frac{\mu_0 M}{B} = \frac{\mu_0 N\mu^2}{3kT} = \frac{C}{T}$$

$\chi = M/H$, $B = \mu_0 H$

The **Curie law**, $C = \mu_0 N\mu^2/3k$ the Curie constant. $1/\chi$ against T is a straight line through the origin. Valid for $\mu B \ll kT$.

Quantum theory of paramagnetism (small-field result)

5 marks

Moment $\mu = g\mu_B J$ with J the total angular momentum, g the Landé factor, $\mu_B = e\hbar/2m$ the Bohr magneton. Only $2J+1$ orientations are allowed: components $-g\mu_B m_J$, $m_J = -J, \dots, J$.

BEFORE YOU START: WHAT THIS USES

$\mu = g\mu_B J$	atomic moment: where an atomic moment comes from; the Landé factor g is derived in the first steps
$\mu_B = e\hbar/2m$	Bohr magneton: where an atomic moment comes from
$m_J = -J, \dots, J$	allowed orientations of the moment (space quantisation): the quantum theory
$E = -\mu \cdot \mathbf{B}$	energy of a moment and the Boltzmann average: Langevin paramagnetism
$\sum m_J^2$	sum of squares: evaluated in the step below

START

$$\mu = -\frac{\mu_B}{\hbar}(\mathbf{L} + 2\mathbf{S})$$

RESULT

$$\chi = \frac{\mu_0 N g^2 J(J+1)\mu_B^2}{3kT} = \frac{\mu_0 N p_{\text{eff}}^2 \mu_B^2}{3kT}$$

LANDÉ FACTOR

$$\mu = -\frac{\mu_B}{\hbar}(\mathbf{L} + 2\mathbf{S})$$

Orbital moment $-\mu_B\mathbf{L}/\hbar$, *spin moment* $-2\mu_B\mathbf{S}/\hbar$

The moment per unit angular momentum is $e/2m$ for orbital motion and twice that for spin ($g_s \approx 2$): see where an atomic moment comes from.

$$\mathbf{J} = \mathbf{L} + \mathbf{S}$$

Total angular momentum

$$\mu_J = -\frac{\mu_B}{\hbar}\,\frac{(\mathbf{L}+2\mathbf{S})\cdot\mathbf{J}}{|\mathbf{J}|}$$

Only the part along $\mathbf{J}$ *survives: the rest turns round* $\mathbf{J}$ *and averages out*

$$(\mathbf{L}+2\mathbf{S})\cdot\mathbf{J} = J^2 + \tfrac{1}{2}\left(J^2 + S^2 - L^2\right)$$

$(\mathbf{L}+2\mathbf{S})\cdot\mathbf{J} = \mathbf{J}^2 + \mathbf{S}\cdot\mathbf{J}$ *and* $\mathbf{S}\cdot\mathbf{J} = \frac{1}{2}(J^2+S^2-L^2)$

Square $\mathbf{L} = \mathbf{J} - \mathbf{S}$: $L^2 = J^2 + S^2 - 2\,\mathbf{J}\cdot\mathbf{S}$.

$$g = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}$$

Replace each squared length by $\hbar^2$ *times (quantum number) (quantum number* $+1$*)*

Then $\mu_J = -g\mu_B|\mathbf{J}|/\hbar$, which is the $\mu = g\mu_B J$ used below. Same formula as in the quantum theory.

ALLOWED ENERGIES

$$E = m_J g \mu_B B$$

Space quantisation: the moment cannot point anywhere, only at $2J+1$ angles.

$$M = N\,\frac{\sum_{m_J}(-m_J g\mu_B)\, e^{-m_J g\mu_B B/kT}}{\sum_{m_J} e^{-m_J g\mu_B B/kT}}$$

Boltzmann average of the component along B

SMALL FIELD

$$M \approx N\,\frac{\sum(-m_J g\mu_B)\left(1-\frac{m_J g\mu_B B}{kT}\right)}{\sum 1}$$

Small field, $u = m_J g\mu_B B/kT \ll 1$: $e^{-u} \approx 1-u$

Top and bottom are both expanded; the bottom sum is just the number of orientations, $2J+1$.

$$M \approx N\,\frac{-g\mu_B\sum m_J + \frac{g^2\mu_B^2 B}{kT}\sum m_J^2}{2J+1}$$

Multiply out

$$\sum_{m_J=-J}^{J} m_J = 0$$

Orientations come in $\pm$ pairs, so the first term vanishes

$$M \approx \frac{Ng^2\mu_B^2 B}{(2J+1)kT}\sum_{m_J} m_J^2$$

simplify

THE SUM OF SQUARES

$$\sum_{m_J=-J}^{J} m_J^2 = 2\sum_{m=1}^{J} m^2 = \frac{J(J+1)(2J+1)}{3}$$

Pairs $\pm m$ give equal terms; $\sum_1^J m^2 = \frac{J(J+1)(2J+1)}{6}$

The sum of squares formula is checked for $J=1$ ($1=\frac{1\cdot2\cdot3}{6}$) and, adding $(J+1)^2$, holds for $J+1$ (induction). Half-integer J gives the same closed form.

$$M = \frac{Ng^2\mu_B^2 B}{3kT}J(J+1)$$

evaluate: $\sum_{-J}^{J} m_J^2 = \frac{1}{3}J(J+1)(2J+1)$

$$\chi = \frac{\mu_0 N p_{\text{eff}}^2\mu_B^2}{3kT}$$

define: $p_{\text{eff}} = g\sqrt{J(J+1)}$

The Curie law again, with $\mu^2 = p_{\text{eff}}^2\mu_B^2$. In strong fields at low T all moments line up and M saturates at $NgJ\mu_B$; for large J the classical Langevin result is recovered. Fits rare-earth salts well; iron-group ions need the crystal field.

Weiss theory: the Curie–Weiss law

8 marks

Weiss: (i) a ferromagnet is made of domains, each spontaneously magnetised; (ii) inside a domain an exchange (molecular) field $B_E = \lambda M$ acts, λ the Weiss constant. Paramagnetic region $T > T_c$.

BEFORE YOU START: WHAT THIS USES

$\mu^2 = g^2J(J+1)\mu_B^2$	square of the atomic moment: quantum paramagnetism
$M = \frac{N\mu^2}{3kT}B$	paramagnetic response to a small field (Curie law): Langevin paramagnetism
$B_E = \lambda M$	Weiss's molecular field, due to exchange: origin of the Weiss field
T_c	Curie temperature: ferromagnetism
$\chi = \mu_0 M/B$	susceptibility: magnetic terminology

START

$$B_{\text{eff}} = B + \lambda M$$

RESULT

$$\chi = \frac{C}{T-T_c}, \quad T_c = \frac{\lambda C}{\mu_0}$$

EFFECTIVE FIELD

$$B_{\text{eff}} = B + \lambda M$$

Each moment feels the applied field plus the molecular field of its neighbours, proportional to the magnetisation. For iron $B_E \approx 1000$ T, far more than any ordinary dipole field: it is the quantum exchange interaction (origin of the Weiss field).

PARAMAGNETIC RESPONSE TO IT

$$M = \frac{N\mu^2}{3kT}(B+\lambda M)$$

Small-field result $M = \frac{N\mu^2}{3kT}B$, with B_{eff} in place of B

Here $\mu^2 = g^2J(J+1)\mu_B^2$ (the quantum form, quantum paramagnetism); the small-field result is the Curie law of Langevin paramagnetism.

SOLVE FOR M

$$M\left(1-\frac{N\mu^2\lambda}{3kT}\right) = \frac{N\mu^2}{3kT}B$$

Collect the M terms

$$M\left(T-\frac{N\mu^2\lambda}{3k}\right) = \frac{N\mu^2}{3k}B$$

multiply: T

Check. What temperature appears in the bracket?

- ✓ $T_c = N\mu^2\lambda/3k$
- $T_c = 3k/N\mu^2\lambda$
- $T_c = \lambda$
- none: it is T alone

$T - N\mu^2\lambda/3k$: define that constant as the Curie temperature.

$$\chi = \frac{\mu_0 M}{B} = \frac{\mu_0 N\mu^2/3k}{T-T_c}$$

divide: $\frac{B}{\mu_0}(T-T_c)$

$$\chi = \frac{C}{T - T_c}$$

substitute: $C = \mu_0 N\mu^2/3k,\ T_c = \lambda C/\mu_0$

The **Curie–Weiss law** (8.64). $1/\chi$ against T is a straight line cutting the T-axis at $+T_c$. At $T = T_c$, $\chi \to \infty$: the solid magnetises itself. Below T_c, $B = 0$ still allows a non-zero solution of $M = f(\lambda M)$: spontaneous magnetisation, largest at 0 K and zero at T_c. The graphical solution is in Weiss theory.

Néel's theory of antiferromagnetism above T_N

5 marks

Two interpenetrating sublattices A and B, equal moments, N atoms each per unit volume. Molecular field constants: α for A–A (and B–B), β for A–B. Both interactions are negative (antiparallel).

BEFORE YOU START: WHAT THIS USES

$A,\ B$	two interpenetrating sublattices with opposite moments: antiferromagnetism
$\alpha,\ \beta$	molecular-field constants for like and unlike neighbours, as in Weiss's λ: Weiss derivation
$M = \frac{N\mu^2}{3kT}B$	each sublattice obeys the Curie law above T_N: Langevin paramagnetism
$C = \mu_0 N\mu^2/3k$	Curie constant: Weiss derivation
$\chi = \mu_0 M/B$	susceptibility: magnetic terminology

START

$$B_A = B - \alpha M_A - \beta M_B$$

RESULT

$$\chi = \frac{2C}{T + \theta}$$

FIELDS AT THE TWO SITES

$$B_A = B - \alpha M_A - \beta M_B$$

The minus signs: the neighbours' fields oppose.
$B_B = B - \beta M_A - \alpha M_B$.

EACH SUBLATTICE IS PARAMAGNETIC

$$M_A = \frac{N\mu^2}{3kT}B_A, \quad M_B = \frac{N\mu^2}{3kT}B_B$$

Above T_N: the Curie result for each sublattice

$$M = M_A + M_B = \frac{N\mu^2}{3kT}[2B - (\alpha + \beta)M]$$

Add, using $B_A + B_B = 2B - (\alpha + \beta)(M_A + M_B)$

$$\chi = \frac{\mu_0 M}{B} = \frac{2\mu_0 N\mu^2/3k}{T + N\mu^2(\alpha + \beta)/3k}$$

Solve for M/B

$$\chi = \frac{2C}{T + \theta}$$

substitute: $C = \mu_0 N\mu^2/3k,\ \theta = N\mu^2(\alpha + \beta)/3k$

Like Curie–Weiss but with $T + \theta$: $1/\chi$ cuts the axis at $-\theta$. Below T_N the sublattices are ordered; the behaviour of χ there (the cusp at T_N) is described in antiferromagnetism. The Néel temperature follows in the steps below.

THE NÉEL TEMPERATURE

$$M_A = \frac{N\mu^2}{3kT_N}(-\alpha M_A + \beta M_A)$$

substitute: $B = 0,\ M_B = -M_A$

Order sets in with no applied field: $B = 0$, and the two sublattices are opposite, $M_B = -M_A$, with M_A still small at T_N so the Curie-law form applies.

$$T_N = \frac{N\mu^2(\beta - \alpha)}{3k}$$

Cancel M_A and solve for the temperature

$$\frac{T_N}{\theta} = \frac{\beta - \alpha}{\beta + \alpha}$$

divide: $\theta = N\mu^2(\alpha + \beta)/3k$

So $T_N < \theta$ when $\alpha > 0$.

The Lorentz local field $E_{\text{loc}} = E + P/3\varepsilon_0$

6 marks

E: macroscopic field in the dielectric, P: polarisation. Carve an imaginary sphere of radius a round the atom. θ: angle from the field direction.

BEFORE YOU START: WHAT THIS USES

$\mathbf{P}$	polarisation, dipole moment per unit volume: polarisation and susceptibility
E	macroscopic (average) field in the dielectric: $E = E_0 + E_p$
$\sigma_b = \mathbf{P}\cdot\hat{\mathbf{n}}$	bound surface charge: derived in the step "Bound charge"
Coulomb	field of a point charge q at distance r is $q/4\pi\varepsilon_0 r^2$
dipole field	$\mathbf{E} = \frac{3(\mathbf{p}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}}{4\pi\varepsilon_0 r^3}$: from Coulomb's law for $+q$ and $-q$

START

$$E_{\text{loc}} = E_1 + E_2 + E_3 + E_4$$

RESULT

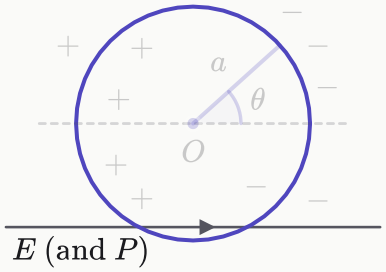
$$E_{\text{loc}} = E + \frac{P}{3\varepsilon_0}$$

SPLIT THE FIELD

$$E_{\text{loc}} = E_1 + E_2 + E_3 + E_4$$

E_1: from the charges on the capacitor plates. E_2: from the bound charge on the outer surface of the dielectric. E_3: from the bound charge on the wall of the sphere. E_4: from the dipoles inside the sphere.

Four contributions to the field at O



$$E_1 + E_2 = E$$

Plates plus outer surface give the macroscopic field

The field of the plates and of the charges on the outer surface together is the average field E inside the dielectric (applied field and the field of the polarisation charges).

BOUND CHARGE

$$\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$$

Bound charge per area = normal component of $\mathbf{P}$, with $\hat{\mathbf{n}}$ pointing out of the dielectric

A thin slab of area A and thickness l has dipole moment PAl (moment per volume times volume). Charges $\pm q$ a distance l apart have moment ql, so $q = PA$ and the charge per area is P. If $\mathbf{P}$ is tilted, only the component along the normal counts.

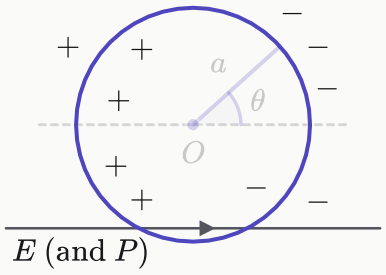
THE SPHERE'S WALL

$$\sigma = -P\cos\theta$$

On the cavity wall the outward normal of the dielectric points into the sphere

Negative charge collects on the side the polarisation points to, positive on the other.

Bound charge on the wall: $\sigma = -P\cos\theta$

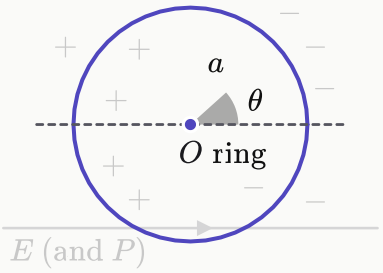


$$dA = 2\pi a \sin\theta \cdot a\, d\theta$$

Area of the ring between θ and $\theta + d\theta$

A ring of radius $a\sin\theta$ and width $a\,d\theta$.

Ring at angle θ, radius $a\sin\theta$

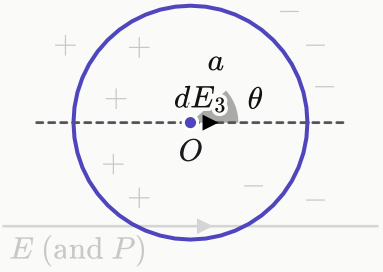


$$dE_3 = \frac{1}{4\pi\varepsilon_0}\frac{P\cos\theta\, dA}{a^2}\cos\theta$$

Coulomb field at O of the ring's charge, along the axis

Every bit of the ring is at distance a; only the component along the field ($\cos\theta$) survives; the sideways parts cancel round the ring.

Only the axial part dE_3 survives



Check. Why is there a second $\cos\theta$?

- ✓ only the component along the axis survives; the rest cancels round the ring
- the charge is $\cos^2\theta$
- the distance is $a\cos\theta$
- it comes from dA

Each element's field points along the radius; symmetric elements on opposite sides of the ring cancel their sideways parts.

$$dE_3 = \frac{P}{2\varepsilon_0}\cos^2\theta\sin\theta\, d\theta$$

substitute: $dA = 2\pi a^2 \sin\theta\, d\theta$

ADD UP ALL RINGS

$$E_3 = \frac{P}{2\varepsilon_0}\int_0^{\pi}\cos^2\theta\sin\theta\, d\theta$$

integrate with respect to θ

$$E_3 = \frac{P}{2\varepsilon_0}\cdot\frac{2}{3} = \frac{P}{3\varepsilon_0}$$

evaluate: $\int_0^\pi \cos^2\theta\sin\theta\,d\theta = \frac{2}{3}$

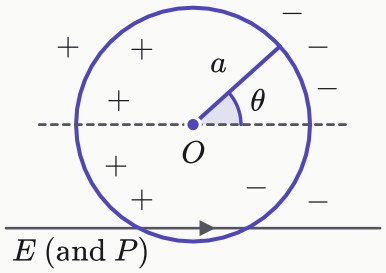
Put $u = \cos\theta$, $du = -\sin\theta\,d\theta$; the limits $0, \pi$ become $1, -1$, and $\int_{-1}^{1} u^2 du = \left[\frac{u^3}{3}\right]_{-1}^{1} = \frac{2}{3}$.

THE DIPOLES INSIDE

$$E_{4,z} = \frac{p}{4\pi\varepsilon_0}\sum_i \frac{3z_i^2 - r_i^2}{r_i^5}$$

Dipoles $\mathbf{p} = p\hat{\mathbf{z}}$ along the field; add their fields at the centre

The field of one dipole $\mathbf{p}$ at position $\mathbf{r}$ from it is $\frac{3(\mathbf{p}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}}{4\pi\varepsilon_0 r^3}$ (Coulomb's law for $+q$ and $-q$). Its z component, with $\mathbf{p} = p\hat{\mathbf{z}}$, is $\frac{p}{4\pi\varepsilon_0}\frac{3z^2 - r^2}{r^5}$.



$$\sum_i \frac{z_i^2}{r_i^5} = \sum_i \frac{x_i^2}{r_i^5} = \sum_i \frac{y_i^2}{r_i^5} = \frac{1}{3}\sum_i \frac{r_i^2}{r_i^5}$$

Cubic symmetry: x, y, z are equivalent

Since $x^2 + y^2 + z^2 = r^2$, each of the three sums is a third of the total.

$$E_4 = \frac{p}{4\pi\varepsilon_0}\left(3\cdot\tfrac{1}{3} - 1\right)\sum_i \frac{1}{r_i^3} = 0$$

substitute

$$E_{\text{loc}} = E + \frac{P}{3\varepsilon_0}$$

substitute

The field acting on an atom is larger than the macroscopic field. It changes with the shape of the specimen and holds for sites of cubic symmetry.

Clausius–Mossotti relation

5 marks

α: polarisability of one atom, $p = \alpha E_{\text{loc}}$. N atoms per unit volume. ε_r: dielectric constant, $D = \varepsilon_0\varepsilon_r E = \varepsilon_0 E + P$.

BEFORE YOU START: WHAT THIS USES

α	polarisability: dipole moment per unit local field: dielectric constant and polarisability
E_{loc}	local field acting on one atom: the Lorentz field derivation
P, D, ε_r	polarisation, displacement and dielectric constant: polarisation and susceptibility
N	atoms per unit volume

START

$$P = N\alpha E_{\text{loc}}$$

RESULT

$$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = \frac{N\alpha}{3\varepsilon_0}$$

ATOMIC AND BULK

$$P = N\alpha E_{\text{loc}}$$

Polarisation = dipole moment per unit volume.

$$P = N\alpha\left(E + \frac{P}{3\varepsilon_0}\right)$$

substitute: $E_{\text{loc}} = E + P/3\varepsilon_0$

$$P = \frac{N\alpha E}{1 - \frac{N\alpha}{3\varepsilon_0}}$$

Collect the P terms

BRING IN THE DIELECTRIC CONSTANT

$$P = \varepsilon_0(\varepsilon_r - 1)E$$

From $D = \varepsilon_0 E + P = \varepsilon_0\varepsilon_r E$

Check. Since $\varepsilon_0\varepsilon_r E = \varepsilon_0 E + P$, what is P?

- ✓ $\varepsilon_0(\varepsilon_r - 1)E$
- $\varepsilon_0\varepsilon_r E$
- $\varepsilon_0(\varepsilon_r + 1)E$
- $(\varepsilon_r - 1)E$

Subtract $\varepsilon_0 E$ from both sides.

$$\varepsilon_r - 1 = \frac{N\alpha/\varepsilon_0}{1 - N\alpha/3\varepsilon_0}$$

Equate the two expressions for P and divide by $\varepsilon_0 E$

$$\varepsilon_r = \frac{1 + 2N\alpha/3\varepsilon_0}{1 - N\alpha/3\varepsilon_0}$$

add: 1

$$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = \frac{N\alpha}{3\varepsilon_0}$$

Solve for $N\alpha/3\varepsilon_0$

A bulk, measurable quantity on the left, an atomic one on the right. For several kinds of atom, $\sum_j N_j\alpha_j/3\varepsilon_0$. At optical frequencies $\varepsilon_r = n^2$ (a light wave in a non-magnetic medium travels at $c/\sqrt{\varepsilon_r}$, so the refractive index is $\sqrt{\varepsilon_r}$; only α_e can follow so fast, frequency dependence): $\frac{n^2-1}{n^2+2} = \frac{N\alpha_e}{3\varepsilon_0}$ (Lorentz–Lorenz).

Electronic polarisability (classical oscillator)

5 marks

An electron bound harmonically to its atom: force constant β, natural frequency $\omega_0 = \sqrt{\beta/m}$. x: its displacement in the local field.

BEFORE YOU START: WHAT THIS USES

β	force constant: the restoring force on a displaced electron is $-\beta x$ (Hooke's law): electronic polarisability
$\omega_0 = \sqrt{\beta/m}$	natural frequency: derived in the first step
$\alpha = p/E_{\text{loc}}$	polarisability: dielectric constant and polarisability
$p = -ex$	dipole moment of the displaced electron: charge × displacement
E_{loc}	local field: Lorentz field

START

$$m\frac{d^2x}{dt^2} = -\beta x \;\Rightarrow\; \omega_0 = \sqrt{\frac{\beta}{m}}$$

RESULT

$$\alpha_e = \frac{e^2}{m(\omega_0^2 - \omega^2)}$$

NATURAL FREQUENCY

$$m\frac{d^2x}{dt^2} = -\beta x \;\Rightarrow\; \omega_0 = \sqrt{\frac{\beta}{m}}$$

Hooke's-law restoring force; try $x \propto \sin \omega_0 t$

Put $x = x_0 \sin \omega t$: $-m\omega^2 = -\beta$, so $\omega_0^2 = \beta/m$. That is why β can be written $m\omega_0^2$ below.

STATIC FIELD

$$-eE_{\text{loc}} = \beta x = m\omega_0^2 x$$

The field's pull is balanced by the restoring force.

$$x = -\frac{eE_{\text{loc}}}{m\omega_0^2}$$

divide: $m\omega_0^2$

$$p = -ex = \frac{e^2}{m\omega_0^2}E_{\text{loc}}$$

multiply: $-e$

$$\alpha_e = \frac{e^2}{m\omega_0^2}$$

$$\alpha = p/E_{\text{loc}}$$

Present in every atom. (A uniform-cloud model in the last steps gives $\alpha_e = 4\pi\varepsilon_0 R^3$.)

ALTERNATING FIELD

$$m\frac{d^2x}{dt^2} + m\omega_0^2 x = -eE_{\text{loc}} \sin \omega t$$

Forced oscillation.

$$m(\omega_0^2 - \omega^2)x_0 = -eE_{\text{loc}}$$

substitute: $x = x_0 \sin \omega t$

Check. With $x = x_0 \sin \omega t$, what is d^2x/dt^2?

- ✓ $-\omega^2 x$
- $\omega^2 x$
- $-\omega x$
- $\omega_0^2 x$

Differentiating $\sin \omega t$ twice brings down $-\omega^2$.

$$\alpha_e = \frac{e^2}{m(\omega_0^2 - \omega^2)}$$

$$p_0 = -ex_0,\ \alpha = p_0/E_{\text{loc}}$$

Frequency dependent, with a resonance at ω_0 (in the ultraviolet). A quantum treatment replaces the one resonance by several: frequency dependence.

UNIFORM-CLOUD MODEL

$$F = \frac{e}{4\pi\varepsilon_0} \cdot \frac{e\,x^3/R^3}{x^2} = \frac{e^2 x}{4\pi\varepsilon_0 R^3}$$

Nucleus displaced by x from the centre of a uniform electron cloud of radius R

By symmetry (Gauss's law) only the cloud charge inside radius x, which is $-e\,x^3/R^3$, pulls, and it acts as if it were at the centre.

$$eE = \frac{e^2 x}{4\pi\varepsilon_0 R^3}$$

Balance with the pull of the field

$$\alpha_e = \frac{ex}{E} = 4\pi\varepsilon_0 R^3$$

$$\alpha = p/E \text{ with } p = ex$$

Proportional to the volume of the atom (electronic polarisability).

Ionic polarisability

3 marks

Positive and negative ions displace relative to each other by Δx; β: force constant of the bond.

BEFORE YOU START: WHAT THIS USES

β	force constant of the ionic bond: ionic polarisability
Δx	relative displacement of the positive and negative ions in the field
$p = e\Delta x$	dipole moment of charges $\pm e$ moved apart by Δx
$\alpha = p/E$	polarisability: dielectric constant and polarisability

START

$$\beta\Delta x = eE$$

RESULT

$$\alpha_i = \frac{e^2}{\beta}$$

BALANCE

$$\beta \Delta x = eE$$

Field force on the ion balanced by the spring-like restoring force.

$$p = e\Delta x = \frac{e^2}{\beta} E$$

multiply: $\frac{e}{\beta}$

$$\alpha_i = \frac{e^2}{\beta}$$

$$\alpha = p/E$$

With $\beta \sim 20\ \mathrm{N\,m^{-1}}$, $\alpha_i \sim 10^{-39}\ \mathrm{F\,m^2}$. Important at low frequencies: NaCl has $\varepsilon_r = 5.6$ static but $n^2 = 2.25$ at optical frequencies.

Dipolar (orientational) polarisability $\alpha_d = p^2/3kT$

5 marks

Molecules with a permanent dipole moment p (e.g. $\mathrm{H_2O}$), N per unit volume. θ: angle between a dipole and $\mathbf{E}$.

BEFORE YOU START: WHAT THIS USES

p	permanent dipole moment of a molecule: dipolar polarisability
$U = -pE\cos\theta$	energy of a dipole in a field: the first two steps (same as for a magnetic moment in Langevin paramagnetism)
$e^{-U/kT}$	Boltzmann factor: Boltzmann tail
$L(x) = \coth x - 1/x$	Langevin function, with its small-x form: Langevin paramagnetism
$\alpha = \bar{p}/E$	polarisability: dielectric constant and polarisability

START

$$\tau = pE\sin\theta$$

RESULT

$$\alpha_d = \frac{p^2}{3kT}$$

ENERGY AND NUMBER

$$\tau = pE\sin\theta$$

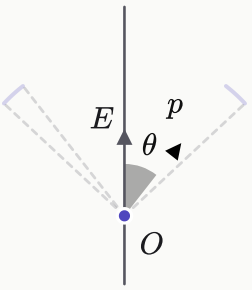
Torque on a dipole: force qE on each charge, arm $\ell\sin\theta$, and $p = q\ell$

The two forces $\pm qE$ form a couple of moment $qE\,\ell\sin\theta = pE\sin\theta$.

$$U = -pE\cos\theta$$

Work against the torque, measured from $\theta = 90°$

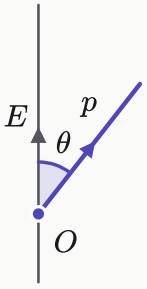
Dipole p at angle θ to E



$$dn \propto e^{pE\cos\theta/kT}\, 2\pi\sin\theta\, d\theta$$

Boltzmann factor × solid angle of the ring

Dipoles between θ and $\theta + d\theta$

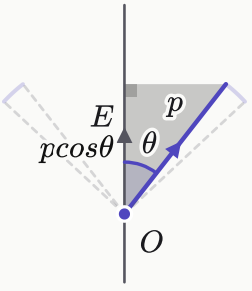


AVERAGE ALONG E

$$\frac{\bar{p}}{p} = \frac{\int_0^\pi \cos\theta\, e^{x\cos\theta}\sin\theta\, d\theta}{\int_0^\pi e^{x\cos\theta}\sin\theta\, d\theta}$$

Average of $\cos\theta$, with $x = pE/kT$

Each dipole gives $p\cos\theta$ along E

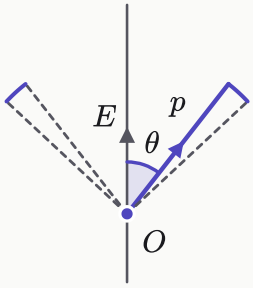


$$\frac{\bar{p}}{p} = \coth x - \frac{1}{x} = L(x)$$

The same integrals as in Langevin paramagnetism, worked out there step by step

$$\bar{p} = \frac{p^2 E}{3kT}$$

Ordinary fields $pE \ll kT$: $L(x) \approx x/3$



$$\alpha_d = \frac{\bar{p}}{E} = \frac{p^2}{3kT}$$

identity

The only polarisability that depends on temperature. Room temperature: $\alpha_d \sim 10^{-39}\ \mathrm{F\,m^2}$, similar to α_e. Total: $\alpha = \alpha_e + \alpha_i + \alpha_d$.

Ferroelectricity: the polarisation catastrophe and $\varepsilon_r \propto 1/(T - T_c)$

5 marks

Clausius–Mossotti with electronic plus ionic polarisabilities α_j of the ions, N_j per unit volume; local field $E + P/3\varepsilon_0$ at every ion.

BEFORE YOU START: WHAT THIS USES

$\alpha_j,\ N_j$	polarisability and concentration of each kind of ion: three ways to polarise
Clausius–Mossotti	$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = \frac{\sum N_j \alpha_j}{3\varepsilon_0}$: Clausius–Mossotti derivation
$E_{\text{loc}} = E + P/3\varepsilon_0$	Lorentz local field
T_c	ferroelectric Curie temperature: ferroelectricity
ξ	a constant (in kelvin) set by how fast the lattice expands: defined in the steps

START

$$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = S, \quad S = \frac{1}{3\varepsilon_0}\sum_j N_j \alpha_j$$

RESULT

$$\varepsilon_r = \frac{3\xi}{T - T_c}$$

DIELECTRIC CONSTANT FROM C–M

$$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = S, \quad S = \frac{1}{3\varepsilon_0}\sum_j N_j \alpha_j$$

Clausius–Mossotti (derivation), with S *for the right-hand side*

$$\varepsilon_r - 1 = S\varepsilon_r + 2S$$

multiply: $\varepsilon_r + 2$

$$\varepsilon_r(1 - S) = 1 + 2S$$

Collect the ε_r *terms*

$$\varepsilon_r = \frac{1 + \frac{2}{3\varepsilon_0}\sum N_j \alpha_j}{1 - \frac{1}{3\varepsilon_0}\sum N_j \alpha_j}$$

divide: $1 - S$

This is the Clausius–Mossotti relation solved for ε_r.

THE CATASTROPHE

$$\sum_j N_j \alpha_j = 3\varepsilon_0$$

The denominator vanishes

Then $\varepsilon_r \to \infty$: a finite polarisation with no applied field. The local field of the polarisation grows faster than the restoring force on the ions; higher-order forces limit the shift.

Check. What happens to ε_r when $\sum N_j \alpha_j$ reaches $3\varepsilon_0$?

- ✓ it becomes infinite: spontaneous polarisation
- it becomes 1
- it becomes zero
- it becomes negative and stays finite

The denominator $1 - \sum N_j \alpha_j / 3\varepsilon_0$ is zero.

JUST ABOVE THE CATASTROPHE

$$\frac{1}{3\varepsilon_0}\sum_j N_j \alpha_j = 1 - s, \quad s \ll 1$$

define

$$\varepsilon_r = \frac{3 - 2s}{s} \approx \frac{3}{s}$$

substitute

With $S = 1 - s$: $1 + 2S = 3 - 2s$ and $1 - S = s$. ε_r is very sensitive to small departures from the critical value.

$$s = \frac{T - T_c}{\xi}$$

Assume s *grows linearly above* T_c *(thermal expansion of the lattice)*

$$\varepsilon_r = \frac{3\xi}{T - T_c}$$

substitute

The Curie–Weiss form seen in the paraelectric state above T_c ($\mathrm{BaTiO_3}$, 380 K).